

Tech Interview Prep:

Core LLM Foundations & Systems Architecture

“Explain how tokenization works, how context window limits affect application architecture, and why Large Language Models exhibit non-deterministic behavior.”

Title & Architectural Roadmap

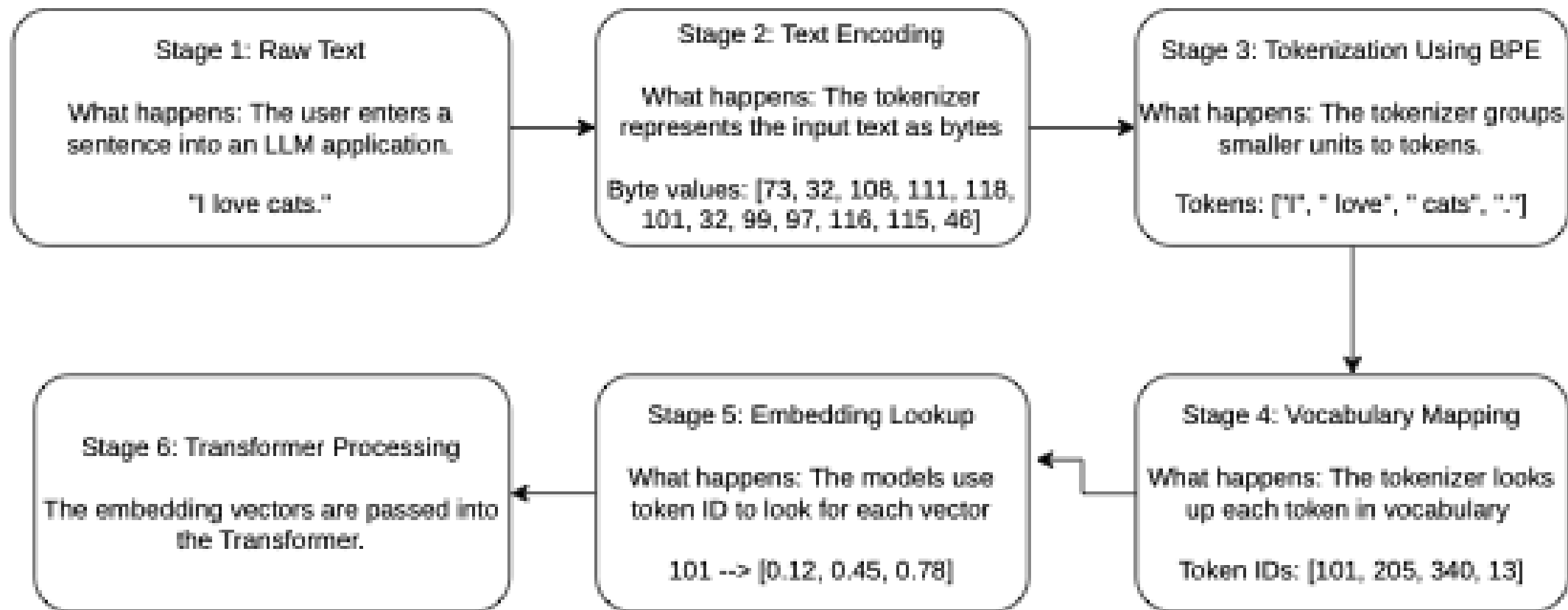
Title: Deconstructing the LLM Runtime: Tokenization, Context Scaling, and Determinism

- 1) **Tokenization** (how text becomes tokens)
- 2) **Context Window** (why context window has limits)
- 3) **Non-Determinism** (why outputs are different for same prompts)

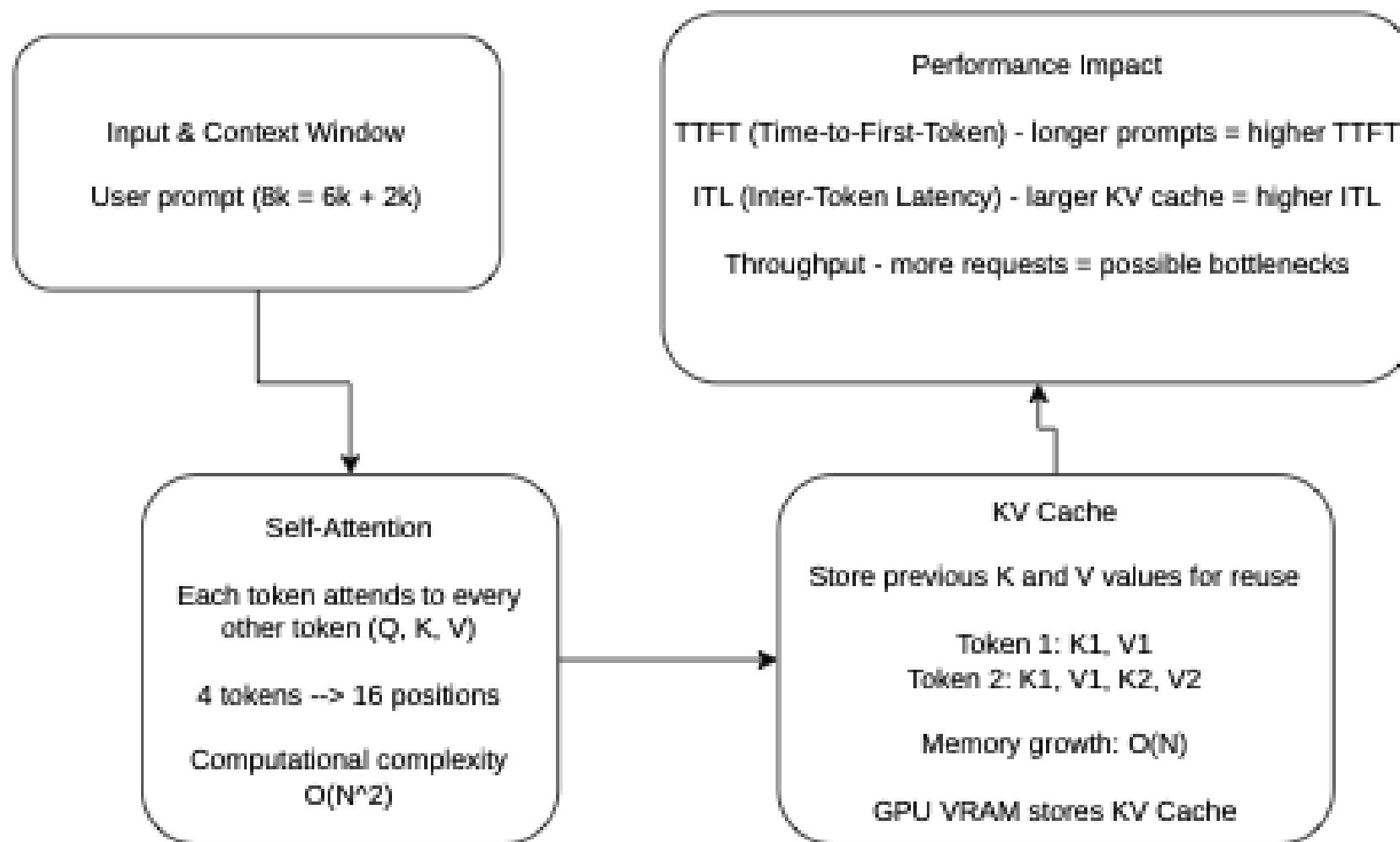
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Deep-Dive: Tokenization Under the Hood



Context Windows: Computational Bounds & Memory Scaling



Application Architecture for Finite & Large Contexts

- Finite Context (Short Context) – Use the model's context window directly
- Large Context (Long Context) – Use an architectural pattern to handle more info

RAG (Retrieval-Augmented Generation)

- Split documents into chunks
- Send selected chunks to the LLM instead of entire documents

Sliding Window

- Keep recent conversation messages
- Remove older messages to manage token limits

Summarization

- Compress older conversations or long documents into shorter summaries
- Preserve important information while reducing context usage

Prefix Caching

- Reuse previously computed KV representations for identical prompt prefixes
- Reduce repeated computation, latency, and cost

Destructuring Non-Determinism: Algorithmic & Hardware Layers

Algorithmic Non-Determinism

Next Token: Cat (50%), Dog (30%), Bird (20%)

- Softmax
- Temperature
- Top-k
- Top-p
- Greedy Decoding ($T=0$)

Hardware Non-Determinism

Meaning: Even without random sampling, hardware and numerical differences can cause output variation

Downstream Mitigation

Meaning: We design applications to handle potentially inconsistent or invalid LLM outputs

- JSON Schema
- Error Handling
- Validation

Summary & Systems Engineering Takeaways

- Tokenization – Manage token budgets to optimize API costs, latency
- Context Windows – Optimize GPU memory and inference performance using KV Caching, RAG, Summarization.
- Non-Determinism - Use JSON Schema validation, and error handling